The Bondi dipole in full numerical relativity: a self-accelerating positive–negative mass binary

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Bondi showed in 1957 that a body of negative active gravitational mass, paired with a positive one, should accelerate together with it indefinitely while the total momentum stays zero. We test this prediction in full 3+1 numerical relativity. The matter consists of two complex scalar fields with identical self-interactions. One of them enters the Einstein equations with the opposite sign, so its inertial and passive masses are positive and its active mass, and with it its Arnowitt–Deser–Misner (ADM) mass, is negative. We construct constraint-satisfying initial data for solitonic boson stars of both signs and run 34 simulations, including same-sign pairs and isolated stars as controls. A pair of stars with equal ADM mass magnitude, released from rest, accelerates as a single object at constant separation, while its total signed matter momentum stays zero to within 1%. The acceleration scales with separation and mass as Bondi's Newtonian law predicts, reverses when the two stars are exchanged, and is absent in every control. We detect no gravitational radiation from the accelerating pair, consistent with the absence of a mass dipole. To our knowledge, the negative-mass star is the first asymptotically flat body of negative ADM mass evolved in numerical relativity. Our simulations cover head-on pairs released from rest and followed up to a few percent of the speed of light.

I. INTRODUCTION

Bondi [1] distinguished three kinds of mass in general relativity: the inertial mass, which resists acceleration; the passive gravitational mass, on which a gravitational field acts; and the active gravitational mass, which sources the field. He showed that a body whose active mass alone is negative is attracted by a positive mass, while the positive mass is pushed away from it. Released together, the two accelerate in the same direction without limit (Fig. 1), and momentum is conserved because the negative body's momentum counts with a negative sign. Bonnor and Swaminarayan [2] found exact solutions for uniformly accelerated pairs of this kind, later generalized in Ref. [3]. Forward [5] showed that the self-accelerating pair conserves both momentum and energy, for equal and unequal mass magnitudes, so objections to negative mass must rest on other grounds [6, 7]. Newtonian N-body simulations have since applied the same sign rules to cosmological structure formation [42, 43].

Nothing in the Einstein equations requires mass to be positive. Positivity of the total mass is a theorem only for matter that satisfies energy conditions [27, 28], and several candidate sources violate them: Casimir energies, quantum fields in curved spacetime and phantom dark energy [8]. Traversable wormholes [9] and warp spacetimes [10] require such matter. The ghost scalar field that holds a wormhole throat open violates the same energy condition as the negative-mass matter evolved here, though with different field equations (Sec. II). A standing objection to all such matter is Bondi's runaway itself, since a pair that accelerates without external force looks like a perpetual-motion machine [6, 7]. That objection has been examined for point particles and prescribed metrics. The existing nonlinear evolutions of matter that violates energy conditions concern single objects: ghost-scalar wormholes in spherical symmetry [15, 30], static stars of phantom fields [31], and the collapse of an Alcubierre warp bubble in 3+1 dimensions [11]. Whether the runaway survives full nonlinear gravity, with finite bodies, backreaction and radiation, can be checked, and we are not aware of a previous 3+1 evolution of a self-gravitating

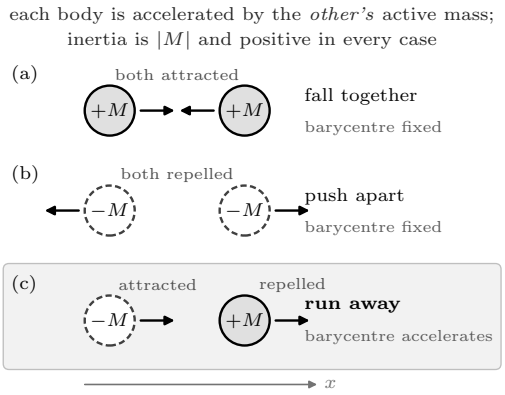


FIG. 1. Bondi's sign rules for two bodies released from rest, after Forward [5]. Every body has positive inertial and passive mass; the active mass of the dashed bodies is negative. (a) Two positive masses attract and (b) two negative masses repel; in both cases the forces are opposite and the barycentre stays at rest. (c) For a positive and a negative mass both forces point the same way, so the pair accelerates while its total momentum stays zero.

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pair of positive and negative active mass with dynamical, constraint-solved matter.

The question also has observational interest. Negative-mass binaries have been proposed as sources of anomalous gravitational-wave signals, such as anti-chirps, dispersal and runaway motion, which detectors could use to exclude them [39]. Nojiri and Odintsov [13] find that a bound positive–negative system can form around a compact mass in a cosmological fluid with a negative cosmological constant. We study the complementary case of an isolated, asymptotically flat pair.

In this paper we realize Bondi's sign structure with two complex scalar fields that obey the same field equation, one of which enters the Einstein equations with a negative sign (Sec. II). Both fields form solitonic boson stars, bound by their own self-interaction, which we construct as constraint-satisfying initial data (Sec. III). We evolve them with `GRTeclyn` [24], the GPU port of `GRChombo` [23] built on AMReX [26], in 34 simulations that include same-sign pairs and isolated stars as controls. Section IV gives the numerical methods, the diagnostics and the five predictions of Bondi's argument that we test; Sec. V gives the results, Sec. VI the gravitational radiation, Sec. VII the systematic checks, and Sec. VIII our conclusions. We use units with $G = c = 1$ and scalar mass $m = 1$. The midpoint X of a pair is the mean of the two stars' barycentres, $\Delta X = X(t) - X(0)$ its displacement, a its acceleration and d the separation of the two stars. Simulations are labelled by the stars they contain, P for a positive-mass star and M for a negative-mass star: PM is a mixed pair, PP and MM are the same-sign controls, and MP is the mixed pair with the two stars exchanged.

II. THE MODEL

The matter must keep positive inertial and passive mass while its active mass is negative, it must form stable, localized stars, and the negative-mass star must be bound by something other than gravity, since its own gravity pushes it apart. The obvious candidates each fail one of these requirements. The negative-mass Schwarzschild spacetime is a naked singularity whose linear perturbations grow exponentially [14], and it contains no matter from which to build initial data. A fluid of negative energy density has no equilibrium: Bonnor's static negative-mass spheres [4] need an interior with no non-gravitational restoring force. A ghost field, whose kinetic term has the wrong sign (a phantom field in the language of cosmology [8]), carries that sign into its own field equation. This inverts its effective potential, which unbinds the star, and it makes the inertial and passive masses negative together with the active mass, as for Forward's negative matter [5]. Such fields collapse or explode in evolutions [15, 30, 45]. We instead reverse the overall sign of one field's Lagrangian, Eq. (2) below, which leaves its field equation unchanged and reverses only its stress-energy tensor.

The stars are solitons: matter bound by its own self-interaction, with gravity a small correction rather than the binding force. We use two independent complex scalar fields, Φ_+ and Φ_-, on one spacetime, with the same sextic potential as in studies of Q-balls and solitonic boson stars [16–18],

$$V(|\Phi|^2) = \tfrac{1}{2}m^2|\Phi|^2 - \tfrac{1}{4}\lambda|\Phi|^4 + \tfrac{1}{6}\mu|\Phi|^6, \tag{1}$$

where m is the scalar mass and $\lambda, \mu > 0$ are the quartic and sextic couplings: the mass term makes the field oscillate, the attractive quartic term lets it clump, and the repulsive sextic term stops the clumping before collapse. Because the two fields share one potential, their matter physics is identical, and the sign introduced next is the only difference between them.

The model is defined by the action

$$S = \int d^4x\,\sqrt{-g}\left[\frac{R}{16\pi} + \mathcal{L}[\Phi_+] - \mathcal{L}[\Phi_-]\right], \tag{2}$$

where g is the determinant of the spacetime metric, R its Ricci scalar, and $\mathcal{L}[\Phi] = -\frac{1}{2}\nabla_\mu\Phi^*\nabla^\mu\Phi - V(|\Phi|^2)$. Varying Eq. (2) with respect to each field gives the same Klein–Gordon equation for both,

$$\Box\,\Phi_\pm = 2\,\frac{\partial V}{\partial|\Phi|^2}\,\Phi_\pm, \tag{3}$$

because the overall sign of a field's Lagrangian cancels from its own Euler–Lagrange equation; the factor 2 comes from the $\frac{1}{2}$ in the kinetic term. Varying with respect to the metric gives

$$G_{\mu\nu} = 8\pi\left(T_{\mu\nu}[\Phi_+] - T_{\mu\nu}[\Phi_-]\right), \tag{4}$$

where $T_{\mu\nu}[\Phi]$ is the standard stress-energy tensor of a complex scalar field: here the sign survives.

Equations (3) and (4) are consistent with each other and with the Bianchi identity $\nabla^\mu G_{\mu\nu} = 0$. Each field's stress-energy tensor is conserved on solutions of its own field equation (3), so the signed sum on the right of Eq. (4) is conserved as well. Reversing the sign of $T_{\mu\nu}[\Phi_-]$ is therefore compatible with conservation of the total source, and it is what the action (2) prescribes. Reversing only the kinetic term of Φ_-, as for a ghost field, would instead change its field equation. The price of the sign is a violation of the null energy condition (NEC), $T_{\mu\nu}k^\mu k^\nu \geq 0$ for every null vector k^μ: the source $-T_{\mu\nu}[\Phi_-]$ violates it wherever $\Phi_- \neq 0$. A field that gravitates with negative energy admits vacuum decay through graviton exchange [32, 33], so we make no claim about the model's quantum consistency and use it as a classical probe of how gravity would act on such matter, as in phantom cosmology [8]. The sign enters the CCZ4 evolution equations only through algebraic source terms, so the principal part of the evolution system is that of CCZ4 with ordinary scalar matter.

The field Φ_- therefore moves like ordinary matter, with positive inertial and passive mass, but sources gravity with a negative sign, so its active mass is negative, and so is the ADM mass of an isolated Φ_- star. We call the stars of Φ_+ and Φ_- positive-mass and negative-mass stars. In the 3+1 split, the Einstein equations are sourced by the energy density $\rho = n^\mu n^\nu T_{\mu\nu}$, the momentum density $S_i = -\gamma_i{}^\mu n^\nu T_{\mu\nu}$ and the stress $S_{ij} = \gamma_i{}^\mu \gamma_j{}^\nu T_{\mu\nu}$, the projections of the total stress-energy tensor seen by observers moving along the unit normal n^μ to the spatial slices, where γ_{ij} is the spatial metric. Each field contributes with its sign $\epsilon_\pm = \pm 1$,

$$\rho = \sum_{s=\pm} \epsilon_s \, \rho[\Phi_s], \qquad S_i = \sum_{s=\pm} \epsilon_s \, S_i[\Phi_s], \tag{5}$$

and likewise for S_{ij}, where each bracket is the standard expression for one complex scalar field.

The action (2) has no term that couples Φ_+ to Φ_-, so stars of different fields interact only through the metric: in the mixed pair, gravity is the only interaction by construction. Two stars of the same field also attract each other directly through the field's self-interaction where they overlap. This short-range attraction does not depend on the sign of the mass; Sec. V E measures it, and the mixed pair cannot have it.

III. INITIAL DATA

A. Isolated stars

A star of this model is a solution in which the field oscillates with a single frequency, $\Phi = \varphi_0(r) e^{-i\omega t}$; the frequency ω labels the family of solutions, with $\omega = m$ the free-field limit. The ratio λ^2/μ sets the lowest frequency a star can have, $\omega_{\min} = \sqrt{1 - 3\lambda^2/16\mu}$ [17]. All our stars use $\lambda^2/\mu = 4.8$ ($\lambda = 10240$, $\mu \simeq 2.18 \times 10^7$, central field amplitude $\simeq 0.019$), so $\omega_{\min} = 0.316$.

A star built in flat space violates the constraints once its own gravity is included, so we solve for each star together with its gravity, as for boson stars [19–21] with a sextic potential, for both signs of the source. We take a conformally flat spatial metric, $\gamma_{ij} = \psi^4 \delta_{ij}$ with conformal factor ψ, maximal slicing, $K = 0$ with K the trace of the extrinsic curvature, and lapse α. Equations (3) and (4) then reduce to ordinary differential equations for $\psi(r)$, $\alpha(r)$ and $\varphi_0(r)$. The negative-mass star solves the same equations with the sign of the matter source reversed in the metric equations only. Its gravity is repulsive: $\psi < 1$ and $\alpha > 1$ in its core, and its ADM mass is negative. We solve the equations by shooting at fixed frequency, bisecting in the central amplitude, and iterate until the frequency measured at infinity, ω/α_∞, matches the requested value to $\sim 10^{-5}$.

Figure 2 shows both families. The reference pair consists of a positive-mass star at $\omega = 0.750$, with $M_+ = +0.014350$, and a negative-mass star at $\omega = 0.7603$, with

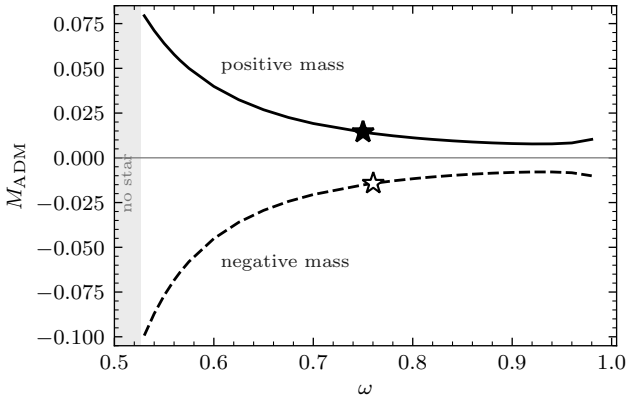


FIG. 2. ADM mass of isolated stars against the oscillation frequency ω, for positive-mass (solid) and negative-mass (dashed) stars at the couplings used here. Star symbols: the two stars of the reference pair, whose $|M_{\rm ADM}|$ agree to 0.4%. At equal ω the negative-mass star is heavier in magnitude, and neither family has a star lighter than about 0.54 of the reference mass.

$M_- = -0.014295$. We call such a pair an equal-mass pair, since $|M_-| = M_+$ to 0.4%. Equal ADM masses are the relevant match because, in Bondi's point-mass limit, each body acts on the other through its exterior field, which the ADM mass sets. We write $\bar{M} = \frac{1}{2}(M_+ + |M_-|)$ for the mean mass of a pair, 0.014322 for the reference pair. The positive-mass star has a root-mean-square (rms) radius of 4.34, the rms distance from its barycentre weighted by w_s of Eq. (6) below, and a compactness $M/R \sim 3\times 10^{-3}$, two orders of magnitude below that of any star near forming a horizon. Four isolated positive-mass stars with $\omega = 0.75$ to 0.90, evolved to $t = 120$, keep their minimum lapse within 1.2×10^{-3} of its initial value. Each star's conjugate momentum is set from the lapse of its own solution, $\Pi = -(\omega/\alpha(r))\,\varphi_0(r)$, so that it starts in its stationary state.

B. The pair

We place the two stars at $x = \pm d_0/2$ about the centre of the domain, where d_0 is the initial separation, and superpose their fields. `GRTresna` [25] then solves the Hamiltonian and momentum constraints for the pair, with each field's sources entering with its sign. Two choices in this step matter for the measurement, and we state them because other studies of mixed-sign matter will meet them.

Both stars on the same slicing. The CTTK method of `GRTresna` [35] can set the initial K algebraically from the local energy density, $K \propto \sqrt{\rho}$, which is imaginary where $\rho < 0$; the negative-mass star therefore requires its maximal-slicing option, $K = 0$, which solves for ψ instead. If the positive-mass star is built with the algebraic option, the two stars are built by different methods, and the positive-mass star starts inside a coherent contraction with $\max|K| \approx 0.1$, four orders of magnitude above

that of the negative-mass star. We therefore use $K = 0$ for both stars. This removes the initial transient in K ($\max|K| \approx 10^{-5}$), and the two constructions then differ only in the sign of ρ.

The same grid spacing for the solve and the evolution. The elliptic solve runs on a domain twice the size of the evolution domain. If its grid spacing differs from that of the evolution, the metric interpolated onto the evolution grid is displaced from the scalar field by a fraction of a cell, so each star starts away from the centre of its own gravitational potential. This produced a drift that did not converge away with resolution and was as large as the effect we measure. We therefore solve on the evolution's grid spacing, so that the cell centres coincide and the metric is copied cell by cell onto the evolution grid, while the scalar field is set on that grid from each star's radial profile. The offset between the centroids of the metric perturbation and of the matter is then below the diagnostic's precision of 10^{-4} in every simulation.

Both stars are given the same internal phase. For the mixed pair the relative phase has no effect, because the two fields do not couple. For two stars of the same field it matters: two Q-balls in phase attract each other through the shared field, and two in antiphase repel [40, 41]. Our same-sign pairs therefore start in the attracting configuration, consistent with the mergers of Sec. V E. Superposed data for two stars of one field also carry phase-dependent constraint errors, which the constraint solve reduces but does not remove [34]; we use these pairs only as controls for the centroid motion.

IV. NUMERICAL METHODS, DIAGNOSTICS AND TESTS

A. Evolution

We evolve the Einstein equations in the conformal and covariant Z4 (CCZ4) formulation [22] with constraint-damping parameters $(\kappa_1, \kappa_2, \kappa_3) = (3, 0, 1)$, 1+log slicing [36] and a Gamma-driver shift [37, 38] with damping parameter $\eta = 1$. Spatial derivatives use fourth-order finite differences, time integration the fourth-order Runge–Kutta method, and we add Kreiss–Oliger dissipation of strength $\sigma = 2$. The CCZ4 variable $\chi = (\det\gamma_{ij})^{-1/3}$, equal to ψ^{-4} on the conformally flat initial data, is below 1 in a positive-mass star and above 1 in a negative-mass star. The time step is $\Delta t = 0.02\,\Delta x$, an order of magnitude below the usual CCZ4 value: it is set by the stability of the sextic self-interaction, and at $\Delta t = 0.2\,\Delta x$ the star disperses. The outer boundaries use Sommerfeld conditions. Outgoing waves of a massive field reflect from boundary conditions designed for massless waves, so an absorbing layer, extra Kreiss–Oliger dissipation that rises with radius, covers $24 \le r \le 32$. With it, the net flux into the domain through its boundary stays below 10^{-7}, even for the same-sign pairs.

All simulations use a uniform grid without mesh refinement, so a change of resolution is a change of the grid spacing alone. The domain is a cube of side $L = 64$ with $N = 128$, 192 or 256 cells per side ($\Delta x = 0.50$, 0.33, 0.25), which puts about 9, 13 and 17 grid points across the rms radius of a star. One simulation doubles the side to $L = 128$ at $N = 256$ ($\Delta x = 0.50$), with the absorbing layer at $48 \le r \le 60$, to extract gravitational waves at larger radii. The reference initial separation is $d_0 = 10$, about twice the radius that contains 90% of a star's weight w_s; other pairs use $d_0 = 8$ to 20. Every pair is released from rest, without a boost, so any motion is dynamical. Most simulations end at $t = 200$; Table I lists the exceptions. Each simulation ran on one NVIDIA H100 GPU; a simulation to $t = 200$ takes about 1, 5 and 15 hours at $N = 128$, 192 and 256.

B. Diagnostics

We track each field in two ways. Its barycentre

$$\bar{x}^{(s)} = \frac{\int x\, w_s\, d^3x}{\int w_s\, d^3x}, \qquad w_s = \left(|\Phi_s|^2 + |\Pi_s|^2\right)^{1/2}, \tag{6}$$

is computed over the whole domain, with Π_s the conjugate momentum of Φ_s. Its core position is the w_s^4-weighted centroid of the cells where w_s exceeds a fixed fraction of its maximum, which ignores the low-amplitude field a star sheds. The main diagnostic is the midpoint of the two stars and its displacement,

$$X = \tfrac{1}{2}\left(\bar{x}^{(+)} + \bar{x}^{(-)}\right), \qquad \Delta X(t) = X(t) - X(0), \tag{7}$$

which Bondi's argument predicts to accelerate and which stays zero by symmetry for the same-sign controls. The two ways of tracking agree on ΔX of the mixed pair to 1%. The separation d is the coordinate distance between the two barycentres; we also measure the proper distance between the two cores along the axis, $\int \sqrt{\gamma_{xx}}\,dx$. The acceleration a is twice the quadratic coefficient of the least-squares fit

$$X(t) = X_0 + v_0\, t + \tfrac{1}{2}\, a\, t^2, \qquad 5 \le t \le t_{\rm end}, \tag{8}$$

which leaves out the first five time units while the gauge adjusts. For point masses with $|M_-| = M_+$, each star is accelerated by the other's active mass, and Bondi's argument predicts

$$a = \frac{\bar{M}}{d^2}. \tag{9}$$

The matter momentum of each field and their signed total are

$$P^{(s)} = \int S_x[\Phi_s]\,\sqrt{\gamma}\,d^3x, \qquad P = P^{(+)} - P^{(-)}, \tag{10}$$

with $\gamma = \det\gamma_{ij}$. We also record the L_2 norms of the Hamiltonian and momentum constraints over the domain, the multipoles of the Newman–Penrose scalar Ψ_4,

which carries outgoing gravitational waves, on spheres of radius R, and the energy conditions. The NEC is violated only where Φ_- is nonzero, and the isolated positive-mass star satisfies it everywhere.

C. Tests

Bondi's argument makes five predictions for a mixed pair with $|M_-| = M_+$, and each has a control: (i) the midpoint X accelerates at a constant rate while the separation d stays constant; (ii) the signed matter momentum P stays zero while the momentum of each field grows; (iii) the acceleration follows Eq. (9), $a \propto d^{-2}$ at fixed mass and $a \propto \bar{M}$ at fixed separation, and each star responds to its partner's mass; (iv) exchanging the two stars reverses the acceleration; (v) pairs without a mass dipole, the same-sign pairs and the isolated stars, do not move their centroids. A numerical artefact would fail at least one of these tests: it need not reverse when the stars are exchanged, need not follow d^{-2} or the partner's mass, and would also appear in the controls. Section V takes the tests in turn.

V. RESULTS

A. The mixed pair accelerates at constant separation

Does the mixed pair accelerate as a single object (test i)? In the reference simulation, PM10 at $N = 256$, the positive-mass star starts at $x = 37$ and the negative-mass star at $x = 27$, and both accelerate toward $+x$ (Figs. 3 and 4). The midpoint moves $\Delta X = 3.00 \pm 0.01$ by $t = 200$, and a single parabola, Eq. (8), fits the whole trajectory with $a = (1.55 \pm 0.01) \times 10^{-4}$; the error bars are the difference between $N = 192$ and 256 (Sec. VII A). The acceleration stays constant in a longer run: in PM10-t400 the acceleration fitted over $133 \le t \le 266$ and over $300 \le t \le 400$ agrees to 2%, and the pair reaches $0.056c$ at $t = 400$. Over $t \le 200$ the separation stays constant to 1%, both as the coordinate distance between the barycentres and as the proper distance between the cores (Table II).

The stars keep their structure while they accelerate. Their peak field amplitude changes by less than 0.3%, the minimum lapse stays above 0.99, and the minimum of χ stays above 0.98, far from collapse, which would drive χ toward zero.

B. The signed momentum stays zero

Does the total momentum stay zero while the stars move (test ii)? The matter momentum of each field grows steadily, to $P^{(\pm)} \simeq 5.3 \times 10^{-4}$ at $t = 200$, while the signed total P of Eq. (10) reaches only 3.8×10^{-6} at $N = 256$, under 1% of either term (Fig. 4d). This residual decreases with resolution, from 3.7×10^{-5} at $N = 128$, so it is a discretization error. The pair therefore moves three units while its total momentum stays zero, as Bondi's argument requires. P is the matter momentum rather than the ADM momentum; we expect the difference, the momentum carried by the gravitational field, to be small at compactness 3×10^{-3}.

C. The acceleration follows $a = \bar{M}/d^2$

Does the acceleration follow Eq. (9) (test iii)? Five pairs with $d_0 = 8$ to 20 test the dependence on separation (Fig. 5a,b). A power law through their accelerations gives $a \propto d^{-2.03 \pm 0.01}$. The ratio $a\,d^2/\bar{M}$ falls from 1.03 at $d_0 = 8$ to 1.00 for $d_0 \ge 12$, as expected for a finite-size correction, which is largest where the two stars overlap most. The signed momentum P at $t = 200$ falls with separation as the acceleration does.

These five simulations ran at $N = 128$, whose acceleration at $d_0 = 10$ lies 7% below the $N = 256$ value. If this offset is common to the five, as the resolution study suggests, it cancels from the exponent and from every ratio in Sec. V F, but not from the absolute values: at $N = 256$, the acceleration of the reference pair lies 8% above $\bar{M}/d^2$, so at a separation of two stellar diameters Eq. (9) holds to about 10%.

Four equal-mass pairs at $d_0 = 20$, where the finite-size correction is 0.2%, test the dependence on mass (Fig. 5c,d). In each pair both stars were rebuilt together to keep $|M_-| = M_+$, since an unequal pair changes its separation (Sec. V F) and Eq. (8) then no longer applies; equal masses need different frequencies for the two fields (Fig. 2). Over a factor 2.46 in $\bar{M}$ the accelerations give $a \propto \bar{M}^{0.97 \pm 0.06}$, and each pair's $a\,d^2/\bar{M}$ lies within 2.4% of 1. These values use the core positions, which ignore the field a star sheds; the barycentres give a steeper exponent, because the lightest pair is the most diffuse and sheds the most field.

D. Exchanging the stars reverses the acceleration

Does the motion reverse when the two stars are exchanged (test iv)? MP10 is PM10 at $N = 128$ with the two stars swapped in place. Its $\Delta X(200)$ and its acceleration are those of PM10 with the opposite sign, to two parts in 10^5. This excludes any asymmetry of the numerical method that does not depend on which star carries the negative mass, such as a biased stencil, an asymmetric domain or a direction-dependent diagnostic. It does not exclude an artefact sourced by the matter, which would reverse as well; the displaced initial data of Sec. III B would have passed this test. The dependence on separation and mass (Secs. V C and V F) and the controls below exclude such artefacts.

TABLE I. The 34 simulations. Labels name the stars, P for a positive-mass and M for a negative-mass star, followed by the initial separation d_0. Unless stated, a simulation uses the stars of the reference pair ($\omega = 0.750$ and 0.7603, $|M_-|/M_+ = 0.996$), the domain $L = 64$, and starts from rest.

Label	Configuration	N	$t_{\rm end}$	Purpose
PM10	mixed pair, $d_0 = 10$	128, 192, 256	200	reference; resolution study
PM8, PM12, PM16, PM20	mixed pair, $d_0 = 8$, 12, 16, 20	128	200	dependence on separation
E0.64, E1.34, E1.57	as PM20, both stars rebuilt with $\|M_-\| = M_+$ (to 0.02%) and $\bar{M}$ equal to 0.64, 1.34 and 1.57 times that of PM20	128	200	dependence on mass
R0.60, R0.80, R1.33	as PM10, one star rebuilt so that $\|M_-\|/M_+ = 0.60$, 0.80, 1.33	128	200	unequal masses
MP10	as PM10, the two stars exchanged	128	200	reversal
PM10-t400	as PM10	128	400	constancy of a
PM10-rec	as PM10, domain recentred on the pair during the run	128	784	separation after $t = 200$
PM10-off	as PM10, pair started 10 units off centre in a fixed domain	128	600	separation after $t = 200$
PM10-tol	as PM10, elliptic-solver tolerance 4.4 times tighter	128	200	initial-data accuracy
PM10-AMR	as PM10, mesh refinement allowed	128	200	grid structure
PM10-η2	as PM10, shift damping $\eta = 2$	128	200	gauge
PM10-L128	as PM10, $L = 128$ ($\Delta x = 0.50$)	256	200	domain size; wave extraction
PP10	two positive-mass stars, $d_0 = 10$	128, 192, 256	200	same-sign control
MM10	two negative-mass stars, $d_0 = 10$	128 (twice), 192	200	same-sign control
P	isolated positive-mass star at the centre of the domain	128	200	isolated control
M, M-t1000	isolated negative-mass star, 5 units off centre	128	200, 1000	isolated control
S0.75–S0.90	isolated positive-mass stars with ω = 0.75, 0.80, 0.85, 0.90	128	120	stationarity of the stars

E. Pairs without a mass dipole stay at rest

Do pairs without a mass dipole stay at rest (test v)? An isolated star moves its core by at most 1.8×10^{-3} on any axis over $t \leq 200$, both at the centre of the domain (P) and five units off centre (M), where no symmetry holds it in place; this is over a thousand times less than ΔX. The barycentre of the off-centre negative-mass star, the quantity that corresponds to the barycentres used for ΔX, moves 1.3×10^{-2}, still about 200 times less.

The same-sign pairs merge, PP10 at $t = 33.6$ and MM10 at $t = 32.8$ (Fig. 6a). These equal merger times are the key result of the controls. Gravity attracts the two positive masses and repels the two negative ones (Fig. 1), so if gravity set the timescale, PP10 would merge while MM10 flew apart. The stars are instead pulled together by the attraction through their shared field (Sec. II), which does not depend on the sign of the mass. Its strength agrees with this: Newtonian free fall from $d = 10$ at these masses takes $t \approx 207$, so merging at $t \approx 33$ requires a force about 35 to 40 times that of gravity. Through infall, merger and the oscillations that follow, the centroids of both same-sign pairs move by less than 8×10^{-4} at every resolution (Fig. 6b), four orders of magnitude below ΔX: neither pair has a mass dipole, and neither centroid moves. The merged objects keep the sign of their curvature, with $\min \chi = 0.979$ for PP10 and $\max \chi = 1.011$ for MM10.

The same comparison shows that the field attraction is absent from the mixed pair, whose two stars are different fields with no coupling. At $d_0 = 10$, where same-field pairs merge within $t \approx 33$, the separation of the mixed pair stays constant to 1% for $t \leq 200$; even a few percent of a force 35 times gravity would have changed it visibly. The mixed pair's acceleration also falls as a power of the separation, whereas an overlap force falls exponentially, and it follows the partner's ADM mass (Sec. V F), on which a force through the field does not depend.

The isolated negative-mass star evolved to $t = 1000$ (M-t1000) reproduces M up to $t = 200$ and then neither collapses nor disperses. Its peak field amplitude decreases linearly, by 0.63% per 100 time units after $t \approx 200$, while its rms radius grows from 4.41 to 4.83 and the minimum of χ stays above 1. The star slowly relaxes outward; we quote this decay rate for the evolved time, not as a statement about stability at later times.

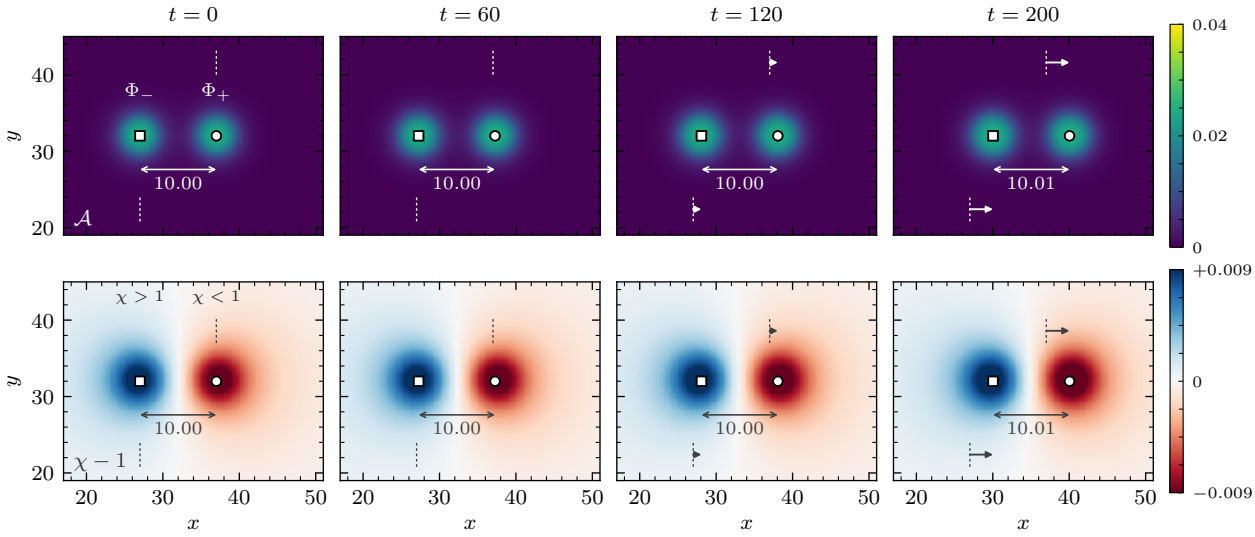


FIG. 3. The reference simulation (PM10, $N = 256$) in the plane $z = 0$ at $t = 0$, 60, 120 and 200. Top: the field weight $\sum_s w_s$ of Eq. (6). Bottom: $\chi - 1$, positive for the negative-mass star (left, blue) and negative for the positive-mass star (right, red). Colour scales are fixed across snapshots. Circles and squares mark the cores of Φ_+ and Φ_-, dashed ticks the initial positions, arrows the displacement, and the number between the stars the separation d. Both stars move toward $+x$ at constant acceleration while d stays constant to 1%. Movie: Video 1, youtu.be/egzPDbz2_B0.

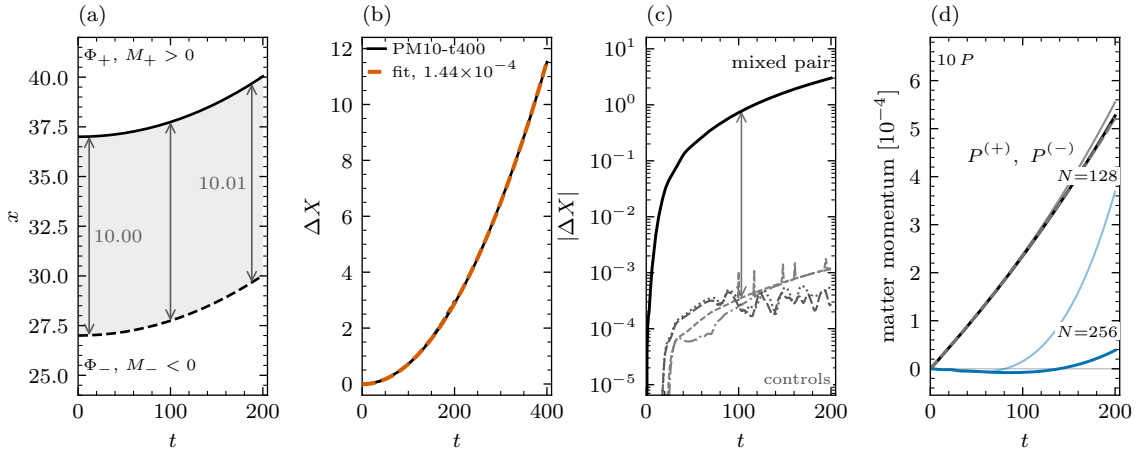


FIG. 4. (a) Core positions of the positive-mass (solid) and negative-mass (dashed) star in the reference simulation, with the separation at three times. (b) ΔX of PM10-t400 (solid) and one constant-acceleration fit, Eq. (8), over $5 \le t \le 400$ (dashed). (c) $|\Delta X|$ of the mixed pair (black) against the core displacement of the isolated stars and the centroid displacement of the same-sign pairs (grey), on a log scale. (d) Matter momentum of each field, $P^{(+)}$ and $P^{(-)}$, and ten times their signed total P (blue; light $N = 128$, dark $N = 256$), Eq. (10). The pair moves three units while P stays below 1% of either field's momentum. Movie: Video 2, youtu.be/zZcr2N12aww.

F. Each star follows its partner's mass

Does each star respond to its partner's mass (test iii)? In Bondi's argument each star is accelerated by the other's active mass, so changing one star's mass must change its partner's acceleration and leave its own unchanged. Three unequal-mass pairs at $d_0 = 10$ test this (Fig. 7). We rebuilt the negative-mass star at $\omega = 0.88$ and 0.804, which gives $|M_-|/M_+ = 0.60$ and 0.80, and the positive-mass star at $\omega = 0.81$, which gives 1.33. Because the separation now changes, we fit each star's acceleration constant C in $\ddot{x} = \pm C/d(t)^2$ with the measured $d(t)$, by regressing the core position $x(t)$ on $\{1, t, I_2(t)\}$ for $t \ge 5$, where I_2 is the double time integral of d^{-2}. We compare only ratios of C between simulations, from which the offset of the $N = 128$ grid cancels. The acceleration of the star whose partner was rebuilt changes in proportion to the partner's mass, to within 0.9 to 3.1%, while that of the star whose partner was unchanged stays within 2.8% of its value in PM10 (Table II). No lighter

TABLE II. Measured results against Bondi's predictions, by simulation (Table I). The pull ratio is the change of a star's fitted acceleration constant C in $\ddot{x} = \pm C/d^2$ when its partner is rebuilt (Sec. V F).

Simulation	Quantity	Measured	Expected
PM10, $N = 256$	$\Delta X(200)$	3.00 ± 0.01	2.86 for $a = \bar{M}/d_0^2$
PM10, $N = 256$ / 192 / 128	a $[10^{-4}]$	1.55 / 1.56 / 1.45	1.43
PM10, $N = 256$	$\lvert P\rvert/P^{(+)}$ at $t = 200$	0.007	0
PM10, $N = 256$	change of d over $t \le 200$: barycentres; proper	-0.9%; $+0.24\%$	0
PM10-t400	a over $133 \le t \le 266$; $300 \le t \le 400$ $[10^{-4}]$	1.45; 1.42	constant
MP10	$\Delta X(200)$; a, ratio to PM10 at $N = 128$	-1.000022; -1.000010	-1
PM8–PM20	exponent of $a(d_0)$	-2.03 ± 0.01	-2
PM8, PM10, PM12, PM16, PM20	$a\, d_0^2/\bar{M}$	1.031, 1.011, 1.003, 1.001, 1.002	1
E0.64, PM20, E1.34, E1.57	exponent of $a(\bar{M})$	0.97 ± 0.06	1
E0.64, PM20, E1.34, E1.57	$a\, d_0^2/\bar{M}$	1.024, 0.997, 0.989, 0.987	1
R0.60, R0.80, R1.33	pull ratio, rebuilt partner	0.618, 0.810, 0.756	0.600, 0.803, 0.747
R0.60, R0.80, R1.33	pull ratio, unchanged partner	1.028, 1.011, 1.001	1
R0.60, R0.80, PM10, R1.33	$\lvert M_-\rvert/M_+$ at which $d(200) = d_0$	0.995 ± 0.03	1
PP10, all N; MM10, all N	largest centroid displacement	7.8×10^{-4}; 5.3×10^{-4}	0
P; M	largest core displacement over $t \le 200$	1.8×10^{-3}; 1.6×10^{-3}	0
PM10-L128	exponent of $\lvert\Psi_4\rvert_{\ell=2}(R)$	-4.8	-1 if radiative
PM10-L128; -tol; -AMR; -η2	change of $\Delta X(200)$ from PM10 at $N = 128$	-4.2%; $+0.0015\%$; $+0.0008\%$; -0.007%	0
M-t1000	decay of the peak field per 100 time units	0.63%	–

partner exists at these couplings: both families have a minimum $|M|$ near 0.54 of the reference mass (Fig. 2).

Unequal masses also change the separation, in the direction Bondi's argument predicts. When the negative-mass star is the heavier, the positive-mass star in front is pushed away more strongly than the negative-mass star behind it is pulled, and the separation grows; when the negative-mass star is the lighter, the separation shrinks. Across the four pairs, the change of the separation at $t = 200$ is monotonic in $|M_-|/M_+$ and crosses zero at 0.995 ± 0.03, against 1 predicted. The error is the spread between ways of measuring the separation (between the barycentres, between the cores, as a proper distance, and averaged over $190 \le t \le 200$). A numerical artefact has no reason to depend on which of the two stars is heavier.

VI. GRAVITATIONAL RADIATION

Does the accelerating pair radiate? The exact solutions for uniformly accelerated pairs are radiative [2, 3], and negative-mass binaries have been proposed as sources of anomalous signals, dipole radiation among them [39]. In our model dipole radiation is absent for the same reason as in ordinary general relativity. The source of gravity is the signed stress-energy tensor, which is conserved (Sec. II), so the time derivative of the signed mass dipole equals the signed momentum P, which stays zero (Sec. V A). Independently, Ψ_4 has spin weight -2, so its multipoles begin at $\ell = 2$. The leading candidate is therefore quadrupole radiation from the accelerating pair. For a pair with zero momentum, the signed mass quadrupole along the axis is $Q \simeq 2\bar{M} d\, X(t)$, so at constant separation and acceleration its second time derivative, $\dddot{Q} = 2\bar{M} a d$, is constant. We measure $\dddot{Q} = 4.5 \times 10^{-5}$, against 4.4×10^{-5} predicted, constant to 7% over $20 \le t \le 190$. The quadrupole luminosity is proportional to $\ddddot{Q}^2$ and vanishes for constant $\dddot{Q}$. From the measured bound $|\ddddot{Q}| \le 6 \times 10^{-8}$, the energy radiated over the run is at most of order 10^{-14}, ten orders of magnitude below the pair's total mass, $M_+ + M_- = 5.5 \times 10^{-5}$.

We extract Ψ_4 on spheres at $R = 16$ to 40 in PM10-L128, whose domain is twice as large (Fig. 8). The $\ell = 2$ amplitude falls as $R^{-4.8}$, and $R\,|\Psi_4|$ drops by a factor 31 across the four spheres, where an outgoing wave would keep it constant. All four spheres therefore lie in the near zone, where Ψ_4 reflects the curvature of the source rather than a radiated flux, and we detect no radiation, in agreement with the estimate above. The radiative regime needs much higher speeds than the $0.03c$ the pair reaches

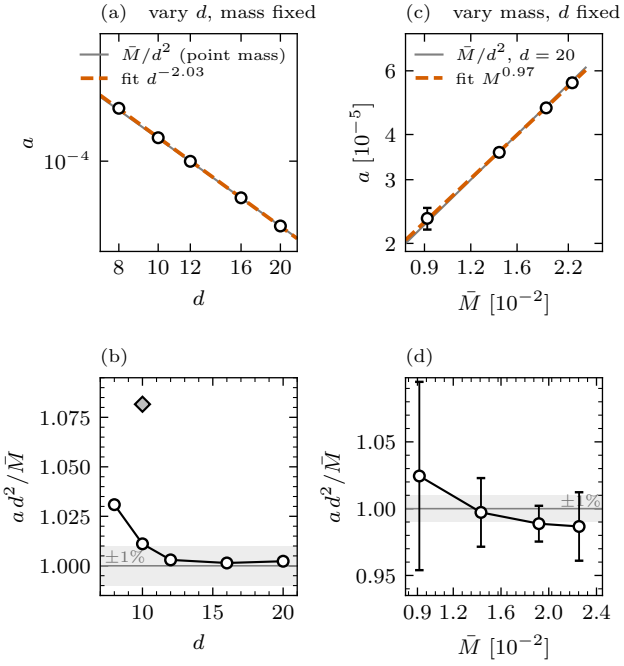


FIG. 5. (a) Acceleration a against initial separation d_0 at $N = 128$ (circles), with a power-law fit (dashed) and $\bar{M}/d^2$ (grey). (b) $a\,d^2/\bar{M}$ against d_0; the diamond is the $N = 256$ value at $d_0 = 10$. (c) a against $\bar{M}$ for the four equal-mass pairs at $d_0 = 20$, with a power-law fit (dashed) and $\bar{M}/d^2$ (grey); (d) their $a\,d^2/\bar{M}$. Error bars: half the spread of fits over four disjoint time windows. The acceleration scales as $d^{-2.03\pm0.01}$ and as $\bar{M}^{0.97\pm0.06}$.

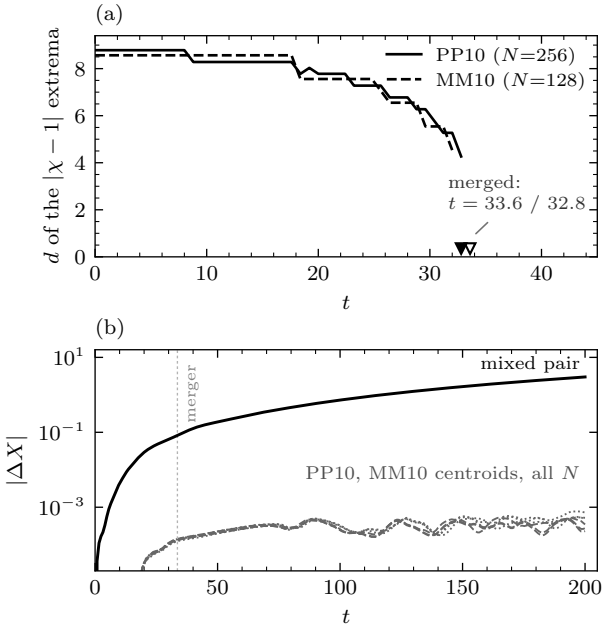


FIG. 6. The same-sign controls. (a) Distance between the two extrema of $|\chi - 1|$ against time for PP10 ($N = 256$, solid) and MM10 ($N = 128$, dashed): both pairs merge, at $t = 33.6$ and 32.8. (b) $|\Delta X|$ of the mixed pair and the centroid displacement of every PP10 and MM10 simulation. Although gravity attracts the stars of PP10 and repels those of MM10, both pairs merge at the same time, through the attraction of their shared field, and their centroids stay below 8×10^{-4}. Movies: Videos 3, youtu.be/jsp-Tv-f68Y, and 4, youtu.be/_J8jfzUk-m4.

by $t = 200$.

VII. CONVERGENCE AND SYSTEMATIC CHECKS

A. Resolution

We evolved PM10 at $N = 128$, 192 and 256 (Fig. 9a). The two highest resolutions agree to 0.4% in $\Delta X(200)$ and to 0.5% in a, and we quote their difference as the error bar: $\Delta X(200) = 3.00 \pm 0.01$ and $a = (1.55 \pm 0.01) \times 10^{-4}$. The lowest resolution lies 4% below in $\Delta X(200)$ and 7% below in a. It is outside the regime in which the three resolutions converge at one order, since the differences between successive resolutions have opposite signs. The simulations of the separation and mass dependences ran at $N = 128$, and we use them only through exponents and ratios, from which this offset cancels. Appendix A gives the constraint norms.

B. Other numerical choices

Simulations that change one setting of PM10 at $N = 128$ test the other numerical choices (Table II). Tightening the elliptic-solver tolerance by a factor 4.4 changes $\Delta X(200)$ by 0.0015%, so the offset of the $N = 128$ grid comes from the evolution, not from the initial-data solve. Allowing mesh refinement changes it by 0.001%; no refined level is created, because $|\chi - 1|$ never reaches the refinement threshold of 0.02. Doubling the shift damping, $\eta = 1 \to 2$, changes the shift by 5% and $\Delta X(200)$ by 0.007%. Doubling the side of the domain, at the same grid spacing, changes $\Delta X(200)$ by -4.2%, so the outer boundary does not drive the motion. Fitting a over the last two thirds of each run instead of $t \geq 5$ changes individual accelerations by 1 to 3% and leaves the exponents, the pull ratios and the zero crossing of Sec. V F unchanged. For the same-sign pairs, whose total mass is $\pm 2M$, the conformally flat outer boundary condition of the elliptic solve is less accurate, and their solves stop at a relative residual near 10^{-5}; their constraint norms stay bounded through the merger, and we use them only as controls of the centroid.

C. Gauge

ΔX and d are coordinate quantities, so we checked the gauge directly. The shift at the two cores is nearly anti-

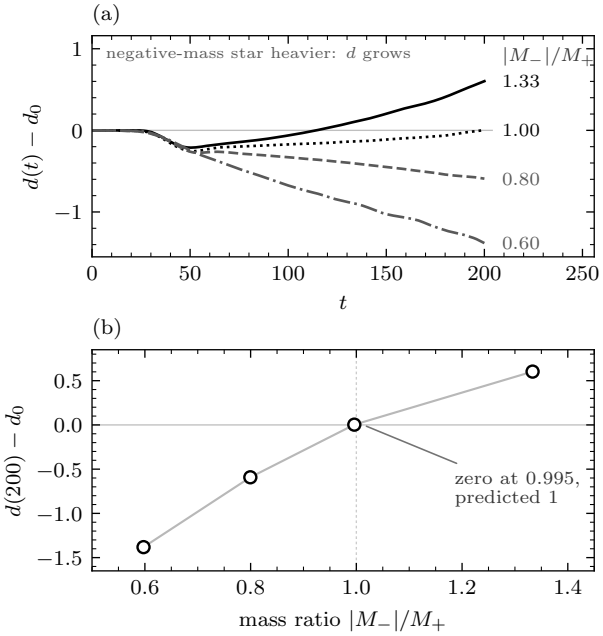


FIG. 7. (a) Change of the separation, $d(t) - d_0$, for the pairs at $d_0 = 10$ with $|M_-|/M_+ = 0.60$, 0.80, 1.00 and 1.33 (R0.60, R0.80, PM10, R1.33); all share a transient near $t \approx 50$, so the change is read at late times. (b) $d(200) - d_0$ against $|M_-|/M_+$. The separation grows when the negative-mass star is the heavier and shrinks when it is the lighter; the change crosses zero at $|M_-|/M_+ = 0.995$, against 1 predicted.

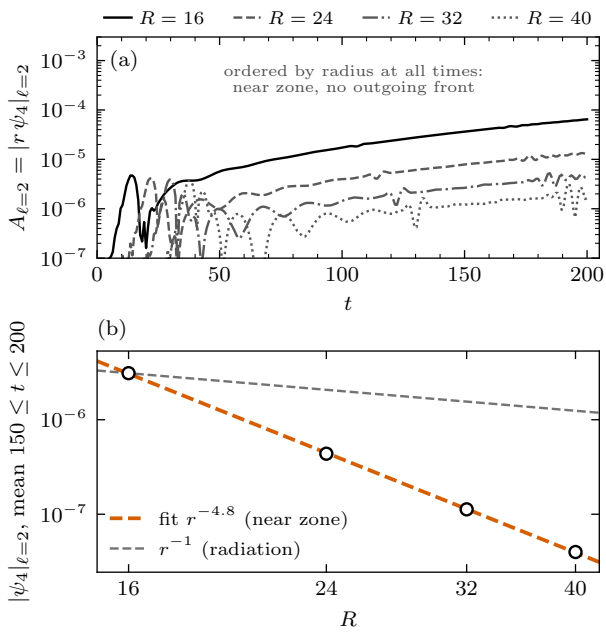


FIG. 8. (a) $\ell = 2$ amplitude of Ψ_4 on spheres of radius $R = 16$, 24, 32 and 40 in PM10-L128 ($L = 128$, $\Delta x = 0.50$). (b) Its mean over $150 \le t \le 200$ against R, with the fit $\propto R^{-4.8}$ (dashed) and the R^{-1} fall-off of radiation (grey). The amplitude falls far faster than radiation would, so all four spheres lie in the near zone and no radiated flux is detected.

symmetric, pointing one way at one star and the other way at the other, so its mean, the only part that could move the pair as a whole, stays below 1.1×10^{-5}, against a midpoint velocity of 3×10^{-2}. Integrated over the run, it accounts for 0.013% of ΔX. The proper distance between the cores, which does not depend on the choice of spatial coordinates along the axis, changes by 0.24% over $t \le 200$. We quote the controls on the cores, because a barycentre over the whole domain also picks up the low-amplitude field that a star sheds, and the mixed pair on both, which agree to 1%.

D. Separation after $t = 200$

At $N = 128$ the separation grows after $t \approx 200$: it reaches 11.8 at $t = 400$ in PM10-t400 and keeps growing in PM10-rec and PM10-off, as the acceleration of the trailing negative-mass star decreases and reverses near $t \approx 530$. With equal masses both stars accelerate equally at every separation, so nothing restores the separation, and any small difference between the two accelerations accumulates. We tested three possible causes: the mass mismatch, the moving domain and the grid resolution. The 0.4% mismatch of the reference pair has the wrong sign, since it would shrink the separation. PM10-rec, whose domain follows the pair, and PM10-off, whose domain stays fixed, agree on the separation to 0.5% up to $t = 500$. Only the resolution changes the effect: the change of the core separation at $t = 200$ falls from 0.107 to 0.029 to 0.013 between $N = 128$ and 256, eightfold for a factor two in grid spacing. We therefore quote pair quantities for $t \le 200$ only, where the change at $N = 256$ is 0.13%.

VIII. CONCLUSIONS

We evolved Bondi's pair of positive and negative active mass as a full initial-value problem in 3+1 numerical relativity, with two complex scalar fields that differ only in the sign with which they source the Einstein equations. Released from rest, a pair with equal $|M_{\rm ADM}|$ accelerates as a single object. Its acceleration is constant to 2% up to $t = 400$, its separation stays constant to 1% up to $t = 200$, and its total signed matter momentum stays zero to better than 1%, with a residual that decreases with resolution. The acceleration follows Bondi's Newtonian law, scaling as $d^{-2.03\pm0.01}$ with separation and as $\bar{M}^{0.97\pm0.06}$ with mass, and each star responds to its partner's mass, so that the separation grows or shrinks according to which star is the heavier. Exchanging the two stars reverses the motion to two parts in 10^5, while the same-sign pairs and the isolated stars, which have no mass dipole, keep their centroids at rest. The same-sign pairs also show that two stars of one field merge through their shared field long before gravity matters,

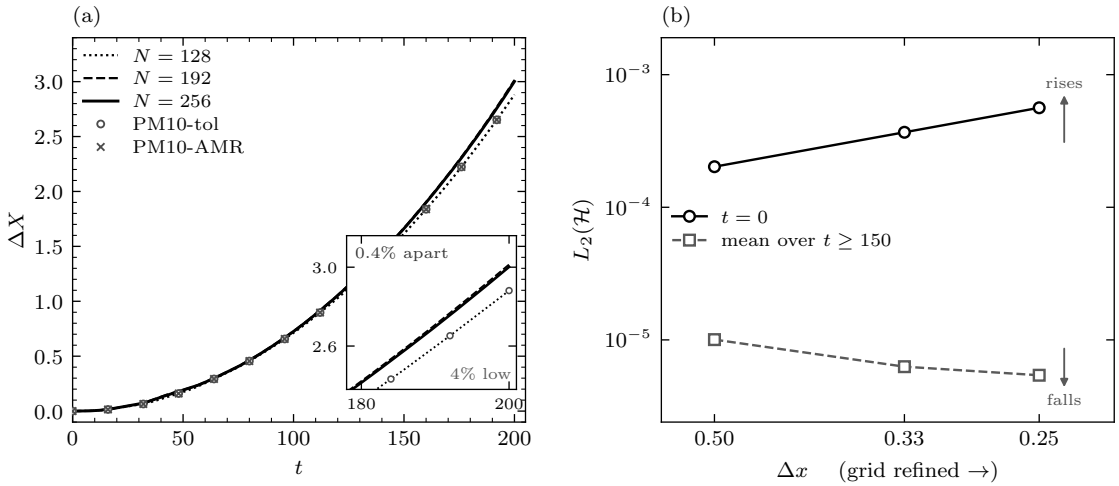


FIG. 9. (a) ΔX of PM10 at $N = 128$, 192 and 256, and of the $N = 128$ simulations with a 4.4 times tighter elliptic-solver tolerance (PM10-tol, circles) and with mesh refinement allowed (PM10-AMR, crosses). Inset: the end of the run. (b) L_2 norm of the Hamiltonian constraint over the whole domain, at $t = 0$ and averaged over $t \geq 150$, against Δx; the $t = 0$ values are dominated by the cells next to the outer boundary (Appendix A). The two highest resolutions agree to 0.4% in $\Delta X(200)$, and $N = 128$ lies 4% below.

an interaction that the mixed pair lacks by construction. We detect no gravitational radiation: the signed dipole cannot change, the second time derivative of the quadrupole is constant, and every extraction sphere lies in the near zone. A star of negative ADM mass exists as constraint-satisfying, asymptotically flat initial data and survives evolution, for 400 time units in a pair and 1000 alone, while it slowly relaxes outward. Static stars of phantom fields [31] and negative-mass bubbles in de Sitter space [29] were known before; negative-mass black holes [12] are a different kind of object.

These results hold for head-on pairs released from rest, in the weak-field regime (compactness 3×10^{-3}), at speeds up to $0.06c$, and over the evolved times. The dependences on separation and mass were measured at $N = 128$ and enter as exponents and ratios, from which the offset of that grid cancels; at $d_0 = 10$, the absolute value of the acceleration agrees with Eq. (9) to about 10%.

Two extensions follow. Releases with a transverse offset would test whether the pair is stable against sideways perturbations. Relativistic speeds would test the question that motivates this configuration: the pair's total mass is only 0.4% of either star's, and no positive-mass theorem bounds the energy it could radiate. Here the speed is limited by the resolution rather than by the length of the run, so that measurement needs finer grids; near $0.3c$ it would provide a concrete source for the signals proposed in Ref. [39]. Nothing here bears on whether such matter exists in nature.

DATA AVAILABILITY

The data that support this study are openly available in the repository of Ref. [44], under `results/bondi-dipole-runaway`, archived on Zenodo (DOI 10.5281/zenodo.23245933). For every simulation they include the time series of all diagnostics, the constraint norms, the Ψ_4 multipoles, the elliptic-solver history and the parameter files; they also include the radial profiles of the stars and the script that regenerates every figure and prints every number quoted here. The initial-data code (`GRTresna` [25]) and the evolution code (`GRTeclyn` [23, 24, 26]) are in the same repository. The Supplemental Material contains movies of the reference simulation (Video 1), of PM10-t400 (Video 2) and of the same-sign controls PP10 and MM10 (Videos 3 and 4).

ACKNOWLEDGMENTS

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Appendix A: Constraint norms

Every constraint solve of a mixed pair converges monotonically and stops below its tolerance, which scales with the grid spacing: the relative residual of the Hamiltonian constraint is 8.6×10^{-6} at $N = 128$ after 13 nonlinear iterations and 8.7×10^{-7} at $N = 256$ after 16. Figure 10 shows the L_2 norms of the Hamiltonian and momentum constraints over the evolution domain, in code units; the domain is mostly vacuum, so norms relative to the size of the matter terms would be dominated by small denominators. The large values at $t = 0$, which grow with

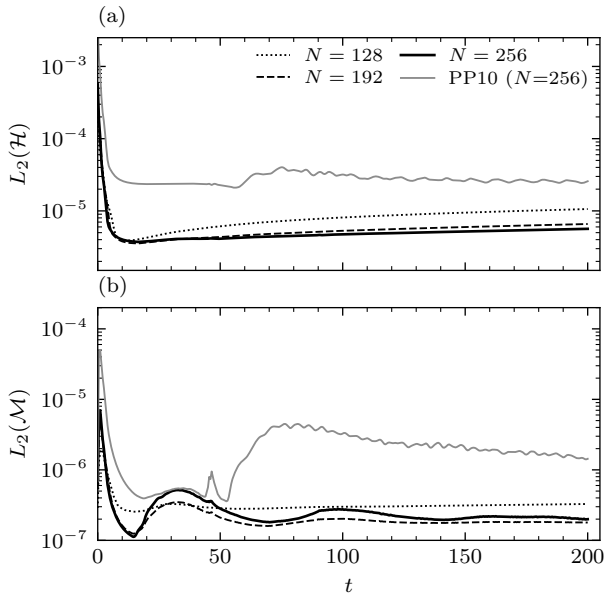


FIG. 10. L_2 norms over the evolution domain of (a) the Hamiltonian and (b) the momentum constraint for PM10 at $N = 128$, 192 and 256 (line styles as in Fig. 9) and for PP10 at $N = 256$ (grey). After a transient from the outer boundary that decays within $t \approx 3$, the norms stay bounded for the whole run.

resolution, come from the cells next to the outer boundary of the evolution domain, which is cut from the larger domain of the elliptic solve: the norm that leaves out the outer boundary layer is a hundred times smaller at $t = 0$ and decreases with resolution. This boundary transient decays by an order of magnitude within $t \approx 3$, before the fit window of Eq. (8) opens. The norms then stay bounded for the rest of the run at every resolution, as they do for PP10 through its merger.

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